Online Retail II UCI Dataset Analysis Project Documentation

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Date: December 26, 2025

Tech Stack: Microsoft Excel, Power BI Desktop, Dataset

GitHub Repository: https://github.com/anesha1/online-retail-ii-powerbi-analysis

Project Overview

This project analyses the Online Retail II UCI dataset from Kaggle (a transactional dataset from a UK-based online gift retailer, spanning December 2009 to December 2011). The goal was to clean the data, explore sales trends, and perform advanced customer segmentation using RFM analysis, all visualized in Power BI Desktop.

Tools used:

- Excel (initial cleaning and exploration)
- Power BI (main transformations, modelling, and dashboard creation)

The final report consists of four pages:

1. **Key Insights** (executive summary of overall business)
2. **Sales Dashboard** (interactive sales exploration)
3. **Customer RFM Segmentation** (detailed RFM technical view)
4. **RFM Insights** (customer segmentation summary with recommendations)

Key metrics (from cleaned data, excluding blank CustomerIDs):

- Total Revenue: £17.33M
- Total Orders: 37K
- Total Quantity Sold: 11M units

- Distinct Customers: 5,852
- Average Order Value: £473.64

Data Source and Initial Cleaning

- **Dataset:** Downloaded from Kaggle
- **Original format:** .xlsx with over 1M rows (exceeded Excel row limit).
- **Cleaning steps in Excel:**
    - Removed rows with blank CustomerID (reduced to customer-focused transactions).
    - Fixed InvoiceDate (converted from serial numbers to proper Date/Time).
    - Renamed columns (e.g., "Price" to "UnitPrice").
    - Converted to Excel Table for better handling.
    - Handled special non-product rows (e.g., "POST", "C2" filtered where needed for product analysis).
    - Saved as "Cleaned Data.xlsx".

MySQL was initially considered for queries but skipped due to import challenges; all work shifted to Power BI for efficiency

Power BI Implementation

Data Import and Transformations (Power Query)

- Imported "Cleaned Data.xlsx" directly.
- Fixed data types: Text for Invoice/StockCode, Decimal for UnitPrice, Date/Time for InvoiceDate, Whole Number for Quantity/CustomerID.
- Added **Revenue** column: Quantity * UnitPrice.
- Other columns: Year, Month from InvoiceDate.

Modeling

- Created key measures:
    - Total Revenue = SUM([Revenue])
    - Total Quantity Sold = SUM([Quantity])
    - Total Orders = DISTINCTCOUNT([Invoice])
    - Distinct Customers = DISTINCTCOUNT([CustomerID])
    - Average Order Value = DIVIDE([Total Revenue], [Total Orders])
    - % Revenue from RFM = DIVIDE(SUM('RFM Table'[Monetary]), [Total Revenue])

Creating RFM Table

For RFM: Created a calculated table "RFM Table" aggregating per CustomerID (Recency days, Frequency orders, Monetary revenue):

dax

```
RFM Table =
VAR SnapshotDate = DATE(2011, 12, 10) + 1  -- One day after last invoice (Dec 9, 2011)
RETURN
SUMMARIZECOLUMNS(
   'sheet 1'[CustomerID],
   "Recency", DATEDIFF(MAX('sheet 1'[InvoiceDate]), SnapshotDate, DAY),
   "Frequency", DISTINCTCOUNT('sheet 1'[Invoice]),
   "Monetary", SUM('sheet 1'[Revenue]))
```

Added new columns to "RFM Table" for quintile scores (1–5):

- Recency Score (5 = most recent):

dax

```
Recency Score =
SWITCH(
   TRUE(),
   'RFM Table'[Recency] <= PERCENTILEX.INC('RFM Table', 'RFM Table'[Recency],
0.2), 5,
   'RFM Table'[Recency] <= PERCENTILEX.INC('RFM Table', 'RFM Table'[Recency],
0.4), 4,
   'RFM Table'[Recency] <= PERCENTILEX.INC('RFM Table', 'RFM Table'[Recency],
0.6), 3,
   'RFM Table'[Recency] <= PERCENTILEX.INC('RFM Table', 'RFM Table'[Recency],
0.8), 2, 1)
```

- Monetary Score (5 = highest spend):

dax

```
Monetary Score =
SWITCH(
   TRUE(),
   'RFM Table'[Monetary] >= PERCENTILEX.INC('RFM Table', 'RFM Table'[Monetary], 0.8),
5,
   'RFM Table'[Monetary] >= PERCENTILEX.INC('RFM Table', 'RFM Table'[Monetary], 0.6),
4,
   'RFM Table'[Monetary] >= PERCENTILEX.INC('RFM Table', 'RFM Table'[Monetary], 0.4),
3,
   'RFM Table'[Monetary] >= PERCENTILEX.INC('RFM Table', 'RFM Table'[Monetary], 0.2),
2, 1)
```

- RFM Score (concatenated, e.g., "555"):

dax

```
RFM Score = FORMAT('RFM Table'[Recency Score], "0") &
         FORMAT('RFM Table'[Frequency Score], "0") &
         FORMAT('RFM Table'[Monetary Score], "0")
```

Additional RFM measures:

- Customers Recency 5 = CALCULATE(DISTINCTCOUNT('RFM Table'[CustomerID]), 'RFM Table'[Recency Score] = 5)
- Customers Frequency 5 = CALCULATE(DISTINCTCOUNT('RFM Table'[CustomerID]), 'RFM Table'[Frequency Score] = 5)
- Customers Monetary 5 = CALCULATE(DISTINCTCOUNT('RFM Table'[CustomerID]), 'RFM Table'[Monetary Score] = 5)
- Champions = CALCULATE(DISTINCTCOUNT('RFM Table'[CustomerID]), 'RFM Table'[RFM Score] = "555")

Relationship: RFM Table[CustomerID] → sheet 1[CustomerID] (one-to-many).

Report Pages and Visuals

1. Key Insights

Executive summary of sales.

- **KPIs** (same as dashboard).
- **Mini visuals:** Top products bar, country map, trend line.
- **Text sections:** Business Overview, Geographic/Temporal/Product Insights, Recommendations (e.g., holiday promotions, international expansion).

2. Sales Dashboard

Interactive exploration page with green theme.

- **KPI Cards** (top): Total Orders (37K), Total Revenue (£17.33M), Total Quantity Sold (11M), Distinct Customers (5,852), Average Order Value (£473.64).
- **Map**: Total Revenue by Country (Top 10) – Highlights UK dominance, Europe/Australia secondary.
- **Line Chart**: Revenue Trend Over Time – Growth to 2010 peak (£8.5M), slight 2011 dip; secondary axis for quantity.
- **Bar Chart**: Top 5 Products by Revenue – Regency Cakestand, White Hanging Heart T-Light Holder, Paper Craft Little Birdie, Jumbo Bag Red Retrospot, Assorted Colour Bird Ornament.
- **Slicers**: Year, Month, Country.

3. RFM Insights (Customer Segmentation Insights)

Narrative summary of RFM.

- **KPIs** (same as RFM page).
- **Mini visuals:** Pie distribution, revenue impact table.
- **Text sections:** Champions, High-Value Core, Potential Loyalists, At-Risk/Hibernating, Lost/Low-Value, Overall Takeaways/Recommendations (e.g., VIP for Champions, win-back emails for At-Risk).

4. Customer RFM Segmentation

Technical deep dive.

- **Cards**: Customers Recency 5 (1,215), Frequency 5 (1,290), Monetary 5 (1,171), Champions (493).
- **Table**: Revenue Impact – RFM Score, Sum/Avg Monetary, % Revenue, Customer Count (Champions drive 48%).
- **Pie Chart**: Customer Distribution by RFM Score (many low-score one-timers).
- **Slicers**: Recency/Frequency/Monetary Scores.

Key Findings and Recommendations

- **Sales**: UK-centric, seasonal (Q4 peaks), vintage gifts dominate.
- **Customers (RFM)**: 80/20 rule strong—Champions (8%) drive 48% revenue. Focus retention on high-scores, nurture mid-tier, low-effort win-back for low-scores.

Troubleshooting

Common issues encountered and solutions:

- **Excel Row Limit Error**:
    - Problem: Opening raw .xlsx triggered "data set too large for Excel grid" warning (dataset >1M rows).
    - Solution: Used Power Query in a new workbook (Get Data > From File) to load full data without sheet limit, or converted to CSV.
- **MySQL Import Failures** (attempted but skipped):
    - Date parsing error ("Incorrect datetime value"): InvoiceDate format mismatch (dd-mm-yyyy vs expected).
    - Decimal error ("Incorrect decimal value"): Comma as decimal separator in UnitPrice.

    - Slow wizard / LOAD DATA LOCAL INFILE restricted (Error 2068).
    - Solution attempted: Import as VARCHAR then convert with STR_TO_DATE/REPLACE; enabled local_infile. Ultimately skipped MySQL in favour of Power BI for faster iteration.
- **DAX Errors in RFM**:
    - NTILE not available in DAX → Replaced with PERCENTILEX.INC + SWITCH for quintiles.
    - Table name mismatch → Ensured consistent use of 'Sheet1' in all references.
- **Power BI Performance**:
    - Large dataset loading slowly → Used Import mode, removed unused columns, applied filters early.

Conclusion

This project successfully transformed the raw Online Retail II UCI dataset into an interactive, insightful Power BI report. Starting from data cleaning in Excel (handling row limits, missing values, and date formatting), the analysis progressed to comprehensive sales visualization and advanced customer segmentation via RFM modeling in Power BI.

Key achievements:

- Built a multi-page dashboard with executive summaries, interactive visuals, and strategic recommendations.
- Implemented RFM analysis from scratch using DAX calculated tables, columns, and measures—revealing the classic 80/20 rule where ~8% of customers (Champions) drive 48% of revenue.
- Demonstrated end-to-end BI skills: data preparation, transformation, modeling, visualization, and storytelling.

Learnings and skills gained:

- Proficiency in Power Query for large datasets.
- Advanced DAX for aggregation and segmentation.

- Best practices in report design (consistent themes, navigation, executive vs detailed pages).
- Decision-making: Pivoting from MySQL to Power BI when tools became blockers.

This report provides actionable insights for the retailer: prioritize loyalty for high-value customers, capitalize on seasonal trends, and explore international growth.

Future enhancements could include cohort analysis, sales forecasting with time intelligence, or integrating live data sources.

Overall, the project showcases a complete business intelligence workflow on a real-world retail dataset.